Artificial Intelligence (AI): Definition & Overview

1. Core Definition

Artificial Intelligence (AI) refers to the simulation of human intelligence processes by machines, especially computer systems. These processes include learning (the acquisition of information and rules for using it), reasoning (using rules to reach approximate or definite conclusions), and self-correction. Essentially, it is a field of computer science dedicated to creating smart machines capable of performing tasks that typically require human intellect.

2. Key Subfields of AI

- **Machine Learning (ML):** A subset of AI that enables systems to automatically learn and improve from experience without being explicitly programmed.
- **Deep Learning (DL):** A specialized branch of ML utilizing multi-layered artificial neural networks to model and solve complex, unstructured data problems.
- **Natural Language Processing (NLP):** The ability of computers to understand, interpret, and manipulate human language (written or spoken).
- **Generative AI:** Deep learning models capable of generating novel, original content, such as text, code, images, or music.

3. Levels of AI Capabilities

- **Artificial Narrow Intelligence (ANI) / Weak AI:** Goal-oriented AI designed to execute specific tasks (e.g., facial recognition, voice assistants, chess engines). All current AI systems belong to this category.
- **Artificial General Intelligence (AGI) / Strong AI:** A hypothetical level of AI where a machine possesses human-level cognitive flexibility, reasoning, and adaptability across diverse domains.
- **Artificial Superintelligence (ASI):** A theoretical future state where AI capabilities surpass the collective intellectual capacity of human beings.

AI Subfield / Type	Primary Function
Machine Learning	Identifies patterns in data to make predictions.
Natural Language Processing	Comprehends, translates, and generates human text/speech.

Narrow AI (ANI)	Excels at performing a single dedicated task efficiently.

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